

The Spatially Resolved Cluster Dynamics (SRCD) Model

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The SRCD model at a glance

MoDELib2 represents defect clusters as **continuum fields**. The model is *reduced*: it resolves only a few cluster classes explicitly.

Mobile clusters (§2.1)

- Small clusters $v, i, 2i, 3i$
- **Fixed** defect content per cluster
- Concentration field c_M solved
- Governed by a reaction–diffusion **PDE**

Immobile clusters (§2.2)

- Large clusters $\rightarrow$ dislocation **loops**
- Tracked by **density** n_l and **content** c_l only
- Size distribution *not* resolved
- Governed by coupled **ODEs**

Interstitial clusters larger than $m_{\max}$ and vacancy clusters larger than $m_{\min}$ are treated as immobile loops. The mobile and immobile fields exchange defects with exact mass conservation.

Notation: two kinds of non-scalar quantities

Quantities carry information in *two different spaces*, distinguished by typeface:

	Species space (indexed by $v, i, 2i, 3i$ or loop family)	Cartesian space (ordinary 3D vectors/tensors)
Typeface	upright sans-serif bold	bold italic
Vectors (lowercase)	$\mathbf{c}_M, \mathbf{J}, \mathbf{S}_{vL}$	$\mathbf{J}_x, \nabla, \sigma$
Matrices (uppercase)	$\mathbf{D}, \mathbf{E}, \mathbf{R}_2$	$\mathbf{D}_x$ (diffusion tensor)

Entries of the species-space flux $\mathbf{J}$ are *themselves* Cartesian vectors $\mathbf{J}_x$, and the divergence $\nabla \cdot$ acts on each entry:

$$\mathbf{J} = [\mathbf{J}_v, \mathbf{J}_i, \mathbf{J}_{2i}, \mathbf{J}_{3i}]^T, \quad \mathbf{J}_x = -\mathbf{D}_x \nabla c_x.$$

Mobile species: definitions

Mobile clusters are set by `mobileSpeciesVector`. Example `Zr_CD4.txt`:

$$\text{mobileSpeciesVector} = -1 \ 1 \ 2 \ 3 \Rightarrow \{v, i, 2i, 3i\}.$$

Collected into the *signed*-size array $\mathbf{m}_M$ with polarity p_k :

$$\mathbf{m}_M = [-1, +1, +2, +3]^T, \quad \mathbf{p}_M = [-1, +1, +1, +1]^T, \quad p_k = \frac{m_k}{|m_k|} \in \{+1, -1\}.$$

- $p_k = +1$: interstitial character; $p_k = -1$: vacancy character; $|m_k|$ = number of point defects.
- Boolean selectors: $\mathbf{M}_i = \frac{1}{2}(1 + \mathbf{p}_M)$, $\mathbf{M}_v = \frac{1}{2}(1 - \mathbf{p}_M)$.
- Mobile concentration field (per-atom, fractional): $\mathbf{c}_M = [c_v, c_i, c_{2i}, c_{3i}]^T$.

$$\text{Point-defect fluxes feeding the loop coupling: } \phi_v = \omega_v c_v, \quad \phi_i = \omega_i (c_i + 2c_{2i} + 3c_{3i}).$$

Mobile master equation (steady-state reaction–diffusion PDE)

$$0 = -\nabla \cdot \mathbf{J} + \mathbf{P}_M + \frac{1}{2}(\mathbf{I} \otimes \mathbf{c}_M^T) \mathbf{R}_2 \mathbf{c}_M + (\mathbf{E} - \mathbf{D}) \mathbf{c}_M + \mathbf{S}_{vL} \quad (1)$$

$-\nabla \cdot \mathbf{J}$	diffusive transport of the four mobile species	
$\mathbf{P}_M = G \varepsilon \mathbf{f}$	cascade production (G = dpa rate, ε survival, $\mathbf{f}$ partition)	(4)
$\frac{1}{2}(\mathbf{I} \otimes \mathbf{c}_M^T) \mathbf{R}_2 \mathbf{c}_M$	second-order reactions (clustering of two mobile species); x -entry = $\frac{1}{2} \mathbf{c}_M^T \mathbf{R}_2^x \mathbf{c}_M$	
$\mathbf{E} \mathbf{c}_M$	thermal dissociation (emission) of mobile clusters	(9)
$-\mathbf{D} \mathbf{c}_M$	annihilation at immobile sinks (network + loops)	(11)
$\mathbf{S}_{vL}$	vacancies returned from dissolving/emitting vacancy loops	(58)

$\mathbf{I}$ is the 4×4 identity; $\otimes$ the Kronecker product. Component-wise, (1) reproduces the 0D ZrMicro rate equations term by term.

Reaction rates, emission, and mobility

Pairwise reaction rate between species m, n (built from summed jump frequencies):

$$\beta_{m,n} = A_{m,n} \Omega^{|M_m M_n|} (D_m + D_n)(r_m + r_n) = (\omega_m + \omega_n), \quad (6,7)$$

consistent with the 0D form $R_{a,b} = (\omega_a + \omega_b)C_a C_b$; the i - v channel carries the recombination factor β_r .

Thermal emission (dissociation) rate from detailed balance:

$$\alpha_{m,n} = \beta_{m,n} \exp\left(-\frac{E_{m,n}^b}{k_B T}\right), \quad E_{m,n}^b = \text{binding energy of the } (m+n) \text{ cluster.} \quad (10)$$

Jump frequency / diffusivity and capture radius:

$$\omega_x = z_c \nu_x \exp\left(-\frac{E_x^m}{k_B T}\right), \quad (D_x)_{ij} = (D_x^0)_{ij} \exp\left(-\frac{E_{ij}^{xm}}{k_B T}\right), \quad r_m = \begin{cases} \left(\frac{3\Omega_0}{4\pi}\right)^{1/3}, & |m_m| = 1 \\ \left(\frac{|m_m|\Omega_0}{\pi b}\right)^{1/2}, & |m_m| > 1. \end{cases} \quad (8)$$

Sinks, loop bias, and boundary conditions

Mobile-to-immobile reaction matrix = network sinks + loop absorption:

$$\mathbf{D} = \text{diag}\{\boldsymbol{\omega} \odot \mathbf{C}_s\} + \mathbf{D}_d \mathbf{K}_L, \quad \mathbf{K}_L = \text{diag}\{\mathbf{k}^2\}, \quad k_m^2 = \sum_{l_L} k_{l_L, m}^2. \quad (11,12)$$

Partial loop sink strengths use per-family bias factors $Z_{l_L}^m$, radii r_{l_L} , densities n_{l_L} (e.g. $k_{a_L, m}^2 = Z_{a_L, \text{tot}}^m 2\pi r_{a_L} n_{a_L}$).
(13–15)

Diffusional-anisotropy-difference (DAD) bias — emerges naturally from the tensorial flux:

$$Z_{a_L, m}^{\text{DAD}, 0} = Z_m^0 \frac{p_m + (p_m)^{-2}}{2}, \quad Z_{c_L, m}^{\text{DAD}, 0} = Z_m^0 p_m, \quad p_m = (D_m^c / D_m^a)^{1/6}. \quad (16)$$

Boundary condition (Dirichlet perfect sink; Robin generalization in the report):

$$\mathbf{c}_M(\mathbf{x}) = \mathbf{c}_{\text{bnd}, \text{eq}}(\mathbf{x}), \quad c_{\text{bnd}, \text{eq}}^k = c_0^k \exp\left(-\frac{m_k t_n \Omega}{k_B T}\right), \quad c_0^k = |m_k| \exp\left(-\frac{E_k^f + m_k q \Omega_k^r}{k_B T}\right). \quad (20-22)$$

Immobile loops: families, fields, and geometry

Four loop families (one vacancy $\langle c \rangle$ loop, three prismatic $\langle a \rangle$ interstitial loops):

$$L = \{vL, ia_1, ia_2, ia_3\}, \quad p_{vL} = -1, \quad p_{ia_1} = p_{ia_2} = p_{ia_3} = +1. \quad (23,24)$$

Each family carries two per-atom fields — **density** $\mathbf{n}_l$ and **content** $\mathbf{c}_l$:

$$\mathbf{n}_l = [n_{vL}, n_{ia_1}, n_{ia_2}, n_{ia_3}]^T, \quad \mathbf{c}_l = [c_{vL}, c_{ia_1}, c_{ia_2}, c_{ia_3}]^T, \quad \bar{m}_k = \frac{c_k}{n_k}, \quad r_k = \ell_k \sqrt{\frac{c_k}{n_k}}. \quad (25-28)$$

Shape transition — vacancy clusters morph from a compact bi-pyramid to a planar loop via a sigmoid $S_k(m_k)$:

$$R_k = (1 - S_k)R_k^{\text{pyr}} + S_k R_k^{\text{loop}}, \quad S_k(m_k) = \frac{1}{1 + \exp\left(-\frac{m_k - m_k^0}{w_k}\right)}. \quad (29,31)$$

The same S_k interpolates the formation-volume tensor Ω_k between pyramid and loop limits. (33,34)

Loop density evolution

$$\frac{d\mathbf{n}_l}{dt} = \dot{\mathbf{N}}_{\text{nuc}} - \mathbf{a}_n - \Delta_N. \quad (35)$$

Nucleation — interstitial loops from clustering + cascade seed; vacancy loops from cascade only:

$$\dot{\mathbf{N}}_{\text{nuc}} = \mathbf{w}(\boldsymbol{\sigma}) \Phi_{\text{clus}} + \mathbf{G}_L \oslash \mathbf{n}_{\text{nuc}}, \quad \Phi_{\text{clus}} = R_{i,3i} + R_{2i,2i}. \quad (37)$$

Stress-biased orientation weights (small-stress limit of SIPN):

$$w_{a_k}(\boldsymbol{\sigma}) = \frac{1}{3}(1 + \chi \hat{\sigma}_{kk}), \quad \sum_k w_{a_k} = 1, \quad \chi_i^k = \frac{\exp(\boldsymbol{\sigma}:\Omega_k/k_B T) l_i^k}{\sum_p \exp(\boldsymbol{\sigma}:\Omega_p/k_B T) l_i^p}. \quad (38)$$

Annealing (vacancy loops only) and **coalescence** (like-loop + loop-network channels):

$$\mathbf{a}_n = \left[\frac{n_{vL}}{\tau_{vL}}, 0, 0, 0 \right]^T, \quad \tau_{vL} = \tau_{vL}^0 e^{E_{vL}^a/k_B T}; \quad \Delta_N^k = (\nu_{LL}^k \phi_{LL}^k + \nu_{LN}^k \phi_{LN}^k) n_k. \quad (36,39)$$

Loop content evolution and defect conservation

$$\frac{d\mathbf{c}_l}{dt} = \dot{\mathbf{G}}_c + \mathbf{P}_G + \mathbf{s}_{\text{nuc}} - \mathbf{A}_c - \mathbf{\Delta}_C. \quad (40)$$

Mass-conserving growth — each family captures its like defect, recombines with the opposite, weighted by DAD+SIPA capture efficiencies $Z_m^k(\sigma)$:

$$\dot{G}_{vL}^c = p_{vL}\pi_{vL}(Z_v^{vL}\phi_v - Z_i^{vL}\phi_i g(r_{vL})), \quad \dot{G}_{a_kL}^c = p_{a_kL}\pi_{a_kL}(Z_i^{a_kL}\phi_i - Z_v^{a_kL}\phi_v). \quad (43-51)$$

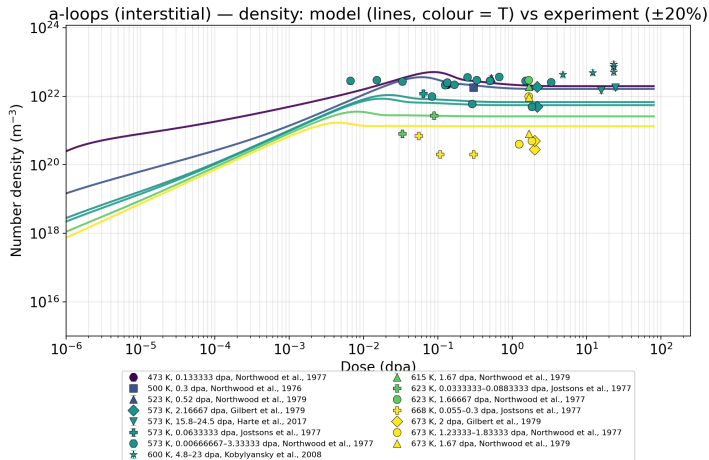
Cascade deposition, nucleation seeding, thermal emission of the mean-size vacancy loop:

$$\mathbf{P}_G = G(\varepsilon_{ia}\mathbf{w}(\sigma) + \varepsilon_{vL}\mathbf{w}'), \quad N_c = 4R_{i,3i} + 2R_{2i,2i} + 3R_{2i,3i}, \quad \alpha_{v,vL}(\bar{m}) = \beta_{v,vL}e^{-E_{v,vL}^b(\bar{m})/k_B T}. \quad (52,53,56)$$

Exact conservation

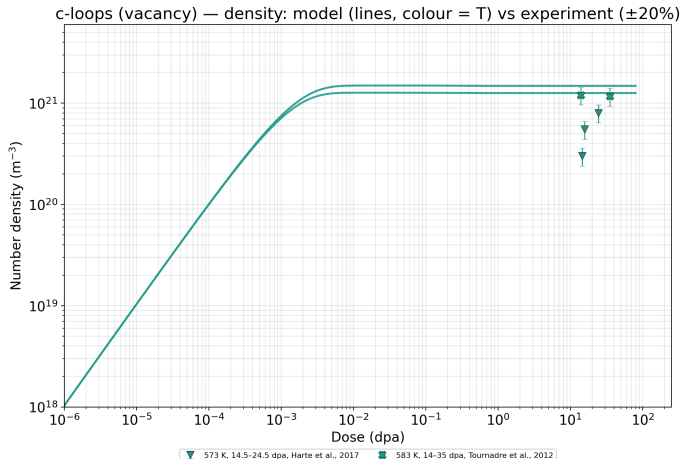
Vacancies released by dissolution ($\mathbf{A}_c$) and emission ($\mathbf{A}_c^{\text{emit}}$) feed the mobile pool via $\mathbf{S}_{vL}$ (Eq. 58), term-by-term identical to the loop debit. Every defect lost to a loop through $\mathbf{D}$ is credited exactly once — **no double counting**.

Calibration: $\langle a \rangle$ -loop density vs. experiment



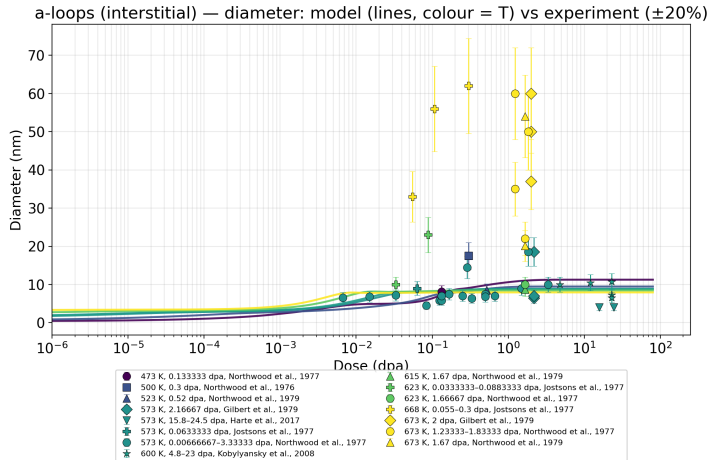
Interstitial $\langle a \rangle$ -loop number density vs. dose, model against the experimental database. The prediction saturates near $4 \times 10^{21} \text{ m}^{-3}$, at the lower edge of the measured Zr band ($\sim 1-3 \times 10^{22} \text{ m}^{-3}$), consistent with the conservative coalescence rate.

Calibration: $\langle c \rangle$ -loop density vs. experiment



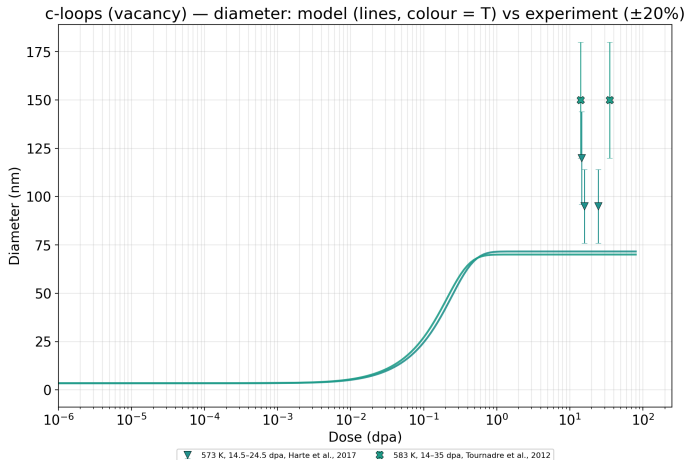
Vacancy $\langle c \rangle$ -loop number density vs. dose against experiment. The predicted density saturates near 10^{21} m^{-3} ; consistent with the anisotropic-bias picture, vacancy loops dominate the stored volume while staying below the interstitial loops in number.

Calibration: $\langle a \rangle$ -loop diameter vs. experiment



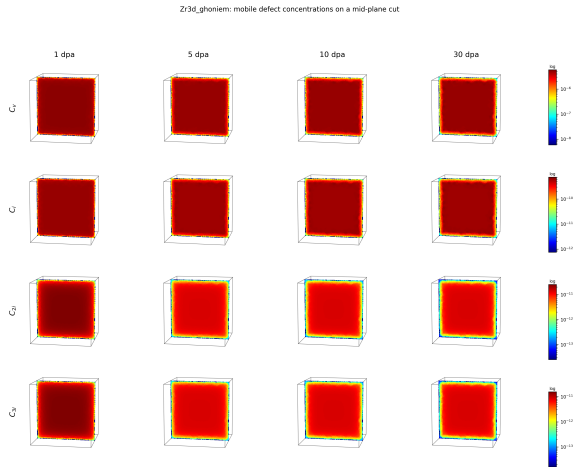
Mean $\langle a \rangle$ -loop diameter vs. dose compared with experiment. The model saturates near 9–10 nm, in good agreement with the bulk of the low- and mid-temperature measurements.

Calibration: $\langle c \rangle$ -loop diameter vs. experiment



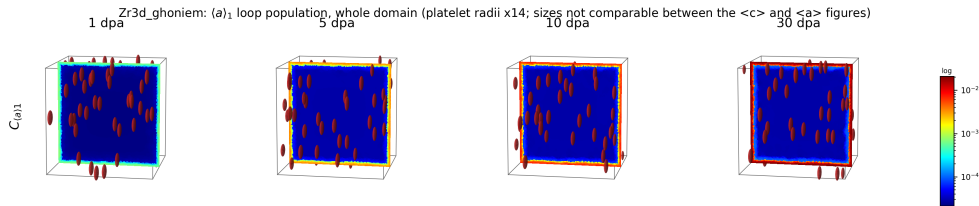
Mean $\langle c \rangle$ -loop diameter vs. dose against experiment. The prediction rises after $\sim 10^{-1}$ dpa toward ~ 70 nm and under-predicts the measured 95–150 nm sizes (Harte, 573 K; Tournadre, 583 K) by ~ 1.5 – $2\times$ — the principal discrepancy motivating the targeted c-loop refit.

Mobile point-defect fields



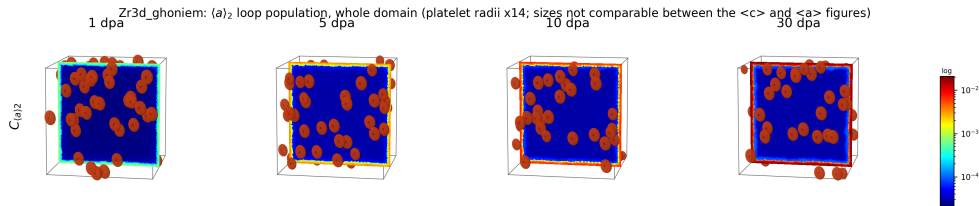
Steady-state concentration fields of the mobile species — vacancies C_v , mono-interstitials C_i , and small interstitial clusters C_{2i} and C_{3i} — on a mid-plane cut at 1, 5, 10, and 30 dpa. All species are nearly uniform in the grain interior and depleted toward the grain boundaries.

$\langle a \rangle$ -loop population: family 1



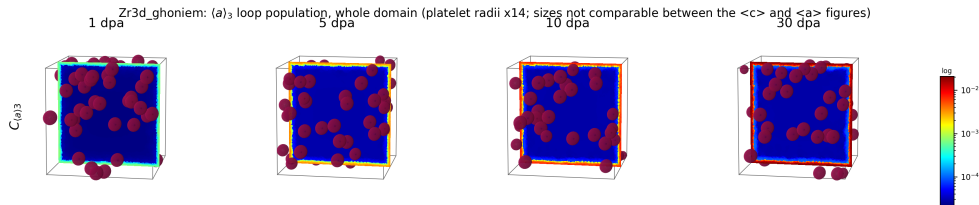
Spatial population of the first $\langle a \rangle$ -loop family $C_{(a)1}$ over the whole domain at 1–30 dpa (platelet markers scaled $\times 14$, log colour scale). Loops nucleate throughout the interior and coarsen with dose while the boundary layer stays depleted.

$\langle a \rangle$ -loop population: family 2



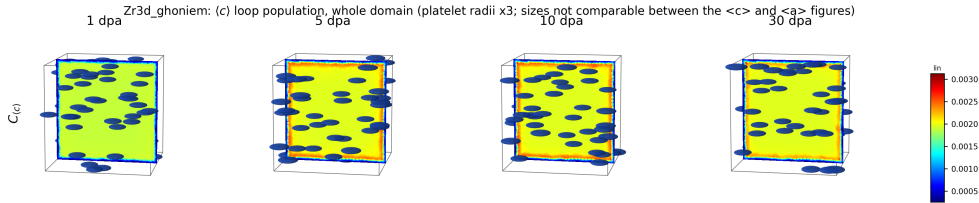
Second $\langle a \rangle$ -loop family $C_{\langle a \rangle 2}$ on the same domain and dose sequence; the growing number and size of markers track progressive $\langle a \rangle$ -loop accumulation.

$\langle a \rangle$ -loop population: family 3



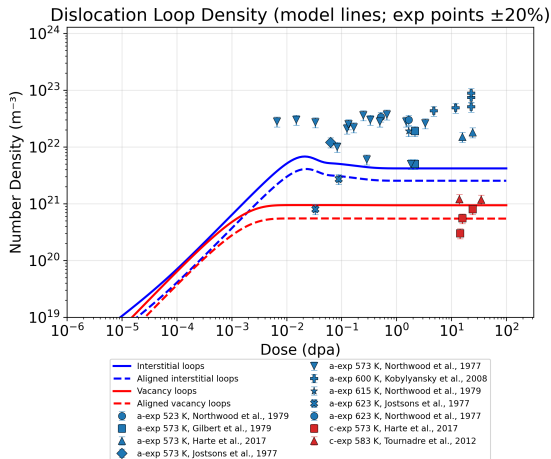
Third $\langle a \rangle$ -loop family $C_{(a)3}$; together the three families represent the full $\langle a \rangle$ -type interstitial loop distribution that feeds the dislocation network.

$\langle c \rangle$ -loop population



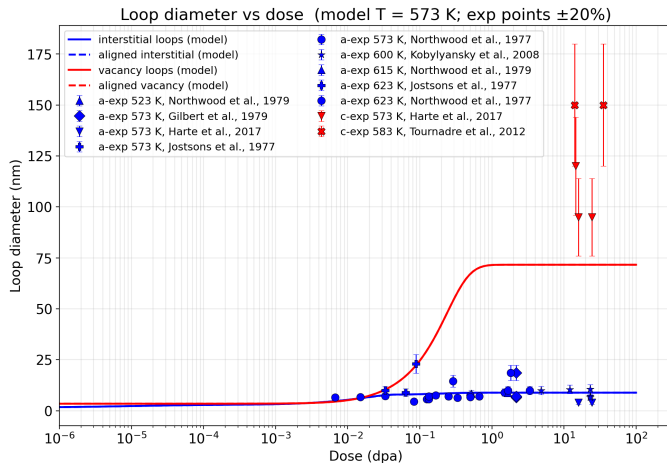
Vacancy $\langle c \rangle$ -loop population $C_{(c)}$ over the domain at 1–30 dpa (linear colour scale, platelet markers $\times 3$). The $\langle c \rangle$ -loops are fewer but individually larger than the $\langle a \rangle$ -loops; marker sizes are not comparable between the $\langle c \rangle$ and $\langle a \rangle$ figures.

Dislocation-loop number density vs. dose



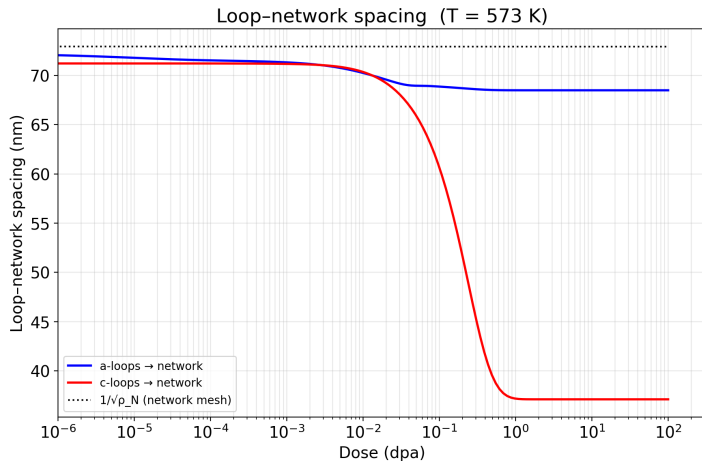
Model loop number densities (interstitial and vacancy, total and network-aligned) compared with the experimental Zr database ($\pm 20\%$). Interstitial $\langle a \rangle$ -loops saturate near a few $\times 10^{21} \text{ m}^{-3}$ and vacancy loops near 10^{21} m^{-3} .

Loop diameter vs. dose



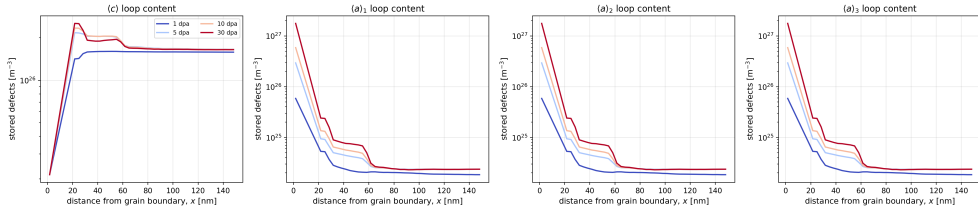
Predicted mean loop diameters at $T = 573$ K against experiment ($\pm 20\%$). Interstitial loops plateau near 9–10 nm, while vacancy $\langle c \rangle$ -loops grow past $\sim 10^{-1}$ dpa toward ~ 70 nm.

Loop-network spacing vs. dose



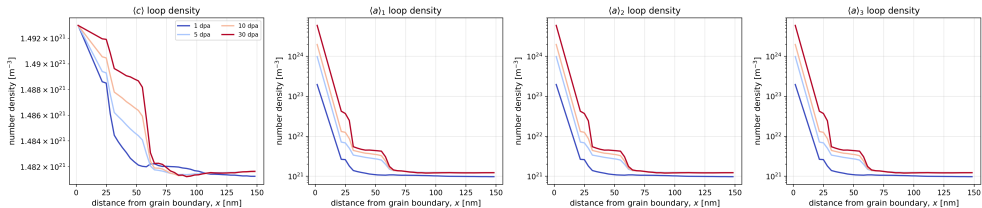
The mean spacing between loops and the dislocation network. The $\langle a \rangle$ -loop spacing stays close to the network mesh size of $1/\sqrt{\rho_N} \approx 70$ nm, whereas the $\langle c \rangle$ -loop spacing collapses to ~ 37 nm above ~ 1 dpa as the $\langle c \rangle$ -loops densify.

Grain boundary content profiles



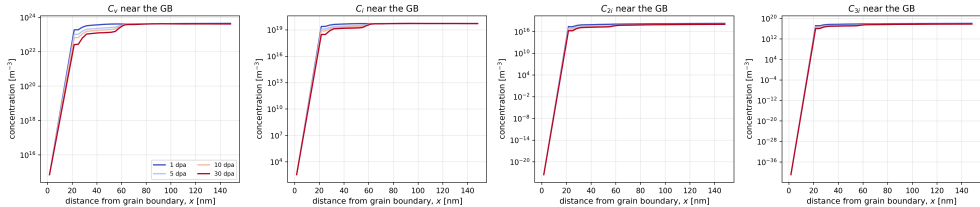
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Grain boundary density profiles



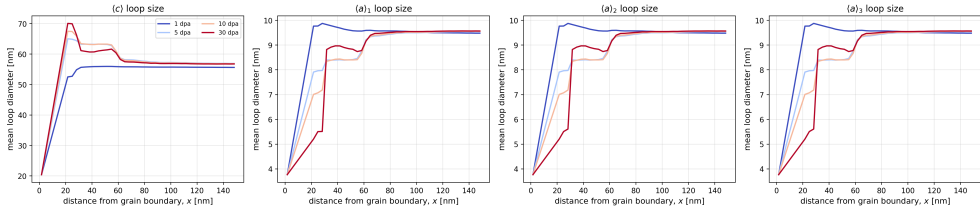
The mean spacing between loops and the dislocation network. The $\langle a \rangle$ -loop spacing stays close to the network mesh size of $1/\sqrt{\rho_N} \approx 70$ nm, whereas the $\langle c \rangle$ -loop spacing collapses to ~ 37 nm above ~ 1 dpa as the $\langle c \rangle$ -loops density.

Grain boundary mobile profiles



The mean spacing between loops and the dislocation network. The $\langle a \rangle$ -loop spacing stays close to the network mesh size of $1/\sqrt{\rho_N} \approx 70$ nm, whereas the $\langle c \rangle$ -loop spacing collapses to ~ 37 nm above ~ 1 dpa as the $\langle c \rangle$ -loops densify.

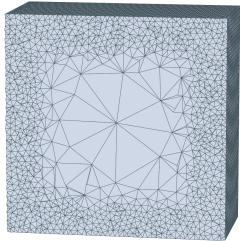
Grain boundary size profiles



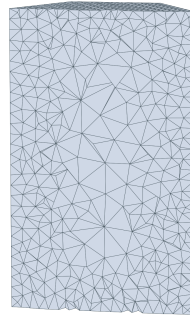
The mean spacing between loops and the dislocation network. The $\langle a \rangle$ -loop spacing stays close to the network mesh size of $1/\sqrt{\rho_N} \approx 70$ nm, whereas the $\langle c \rangle$ -loop spacing collapses to ~ 37 nm above ~ 1 dpa as the $\langle c \rangle$ -loops densify.

Mesh Dependence Studies

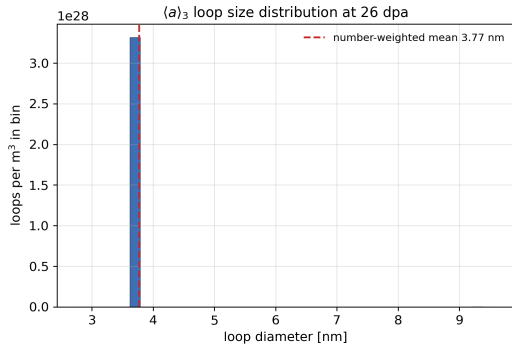
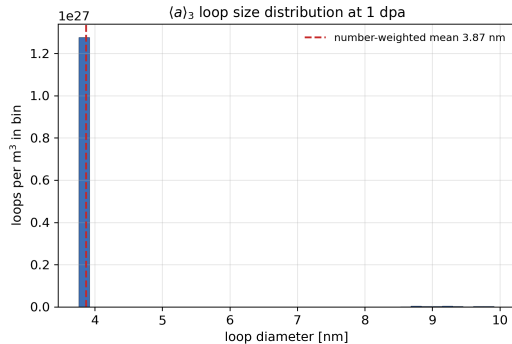
281,218 tets — 29 nm in the 150 nm layer, 147 nm interior (cut plane facing the viewer)



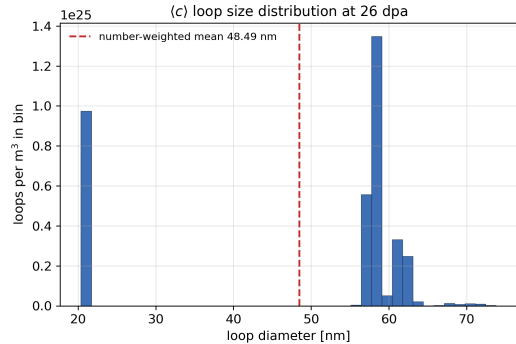
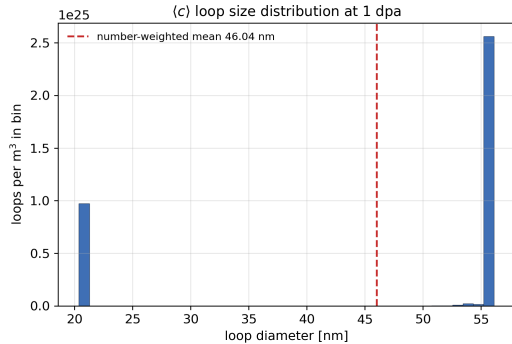
Hexagonal Zr single crystal, 400 nm wide x 653 nm tall — 20,726 tets (cut normal to y)



a-Loop Size Distributions

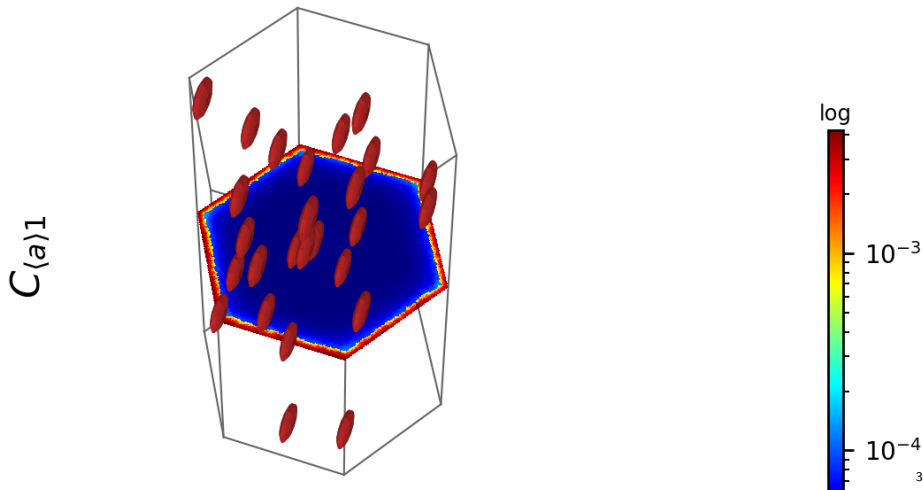


c-Loop Size Distributions



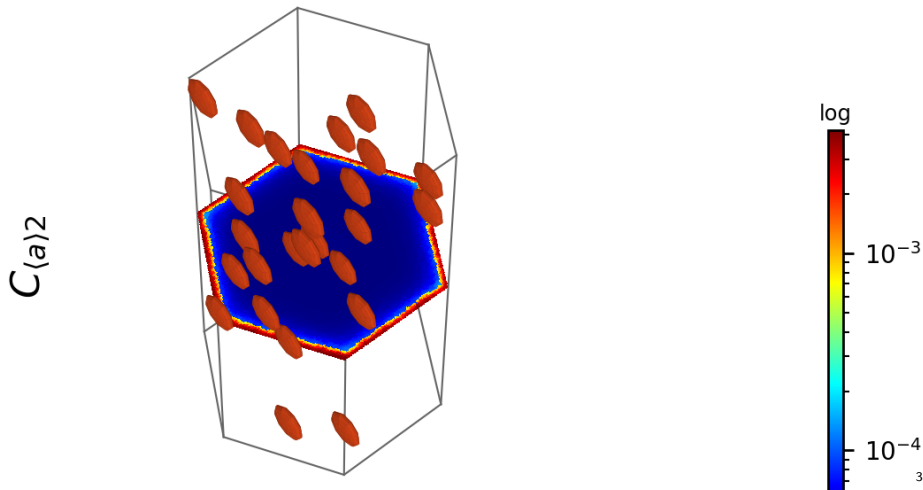
Hexagonal Single Crystal - Unstressed

$\langle a \rangle_1$ loop population at 6 dpa (platelet radii x7, not to scale)



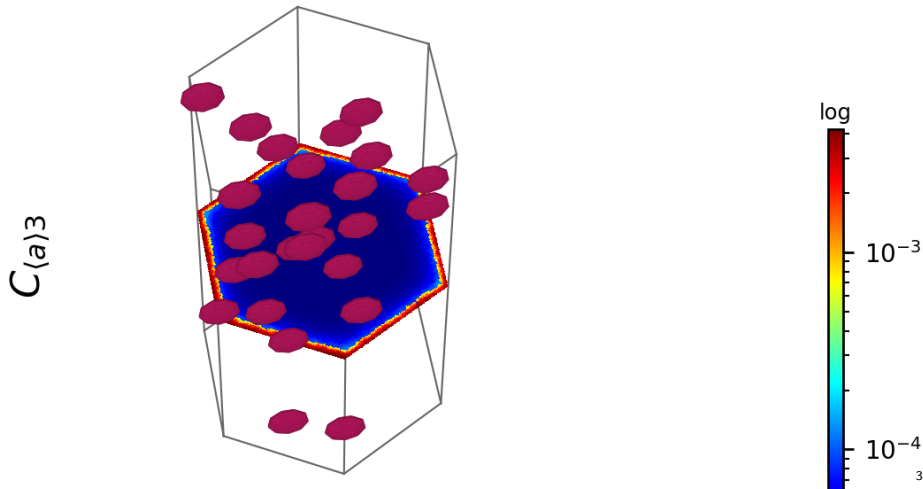
Hexagonal Single Crystal - Unstressed

$\langle a \rangle_2$ loop population at 6 dpa (platelet radii x7, not to scale)



Hexagonal Single Crystal - Unstressed

$\langle a \rangle_3$ loop population at 6 dpa (platelet radii x7, not to scale)



Hexagonal Single Crystal - Unstressed

$\langle c \rangle$ loop population at 6 dpa (platelet radii x1.5, not to scale)

